

# Adjacent Vertices Fault-tolerance Fanability of Hypercube \*

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## Abstract

In this paper, we introduce the concepts of fault tolerant fanability. We show that the  $n$ -dimensional hypercube  $Q_n$  are  $f$ -adjacent and  $l$  edges fault tolerant  $(n - f - l)^*$ -fanable for  $n \geq 3$ ,  $f + l \leq n - 2$  and  $l \geq 1$ .

## 1 Introduction

The hypercube network is one of the most popular interconnection networks. It has many attractive properties, such as regularity, symmetry, small degree and diameter, maximum fault tolerance, easy routing algorithms.

Network topology is usually represented by a graph where vertices represent processors and edges represent links between processors. A bipartite graph  $G = (V_0 \cup V_1, E)$  is a graph such that  $V_0 \cap V_1 = \emptyset$  and  $\forall e = (u, v) \in E, u \in V_0, v \in V_1$ . Let  $V_0(V_1)$  be the set of black(white) vertices. An  $n$ -dimensional hypercube  $Q_n$  is a bipartite graph with  $2^n$  vertices. Each vertex is denoted as an  $n$ -bit binary string. Two vertices of hypercube are adjacent if and only if their binary string representations differ exactly in one bit position.

A path is a sequence of adjacent vertices, denoted as  $\langle v_1, v_2, \dots, v_n \rangle$ , where all the vertices except  $v_1$  and  $v_n$  are distinct. A Hamiltonian path is a path that visits every vertex exactly once. A bipartite graph  $G = (V_0 \cup V_1, E)$  is Hamiltonian laceable if there exists a Hamiltonian path between each pair of vertices  $u$  and  $v$  for  $u \in V_0$  and  $v \in V_1$ . A Hamiltonian laceable graph  $G = (V_0 \cup V_1, E)$  is hyper-Hamiltonian laceable if  $\forall v \in V_0(V_1)$ , there exists a Hamiltonian path of  $G - \{v\}$  between each pair of vertices of  $V_1(V_0)$ . A graph  $G$  is  $k$

edges Hamiltonian laceable if  $G - F$  is Hamiltonian laceable  $\forall F \subset E$  and  $|F| = k$ . A graph  $G$  is  $k$  edges hyper-Hamiltonian laceable if  $G - F$  is hyper-Hamiltonian laceable  $\forall F \subset E$  and  $|F| = k$ . In [5], Tsai et al. showed that  $Q_n$  is  $(n - 2)$  edges Hamiltonian laceable and  $(n - 3)$  edges hyper-Hamiltonian laceable.

The following concepts are introduced in [1]. A  $k$ -container  $C(s, t)$  of a graph  $G$  is a set of  $k$  internal vertex disjoint paths between  $s$  and  $t$ . Let  $V(C(s, t))$  be the set of vertices incident with some paths in  $C(s, t)$ . A  $k^*$ -container  $C(s, t)$  is a  $k$ -container that  $V(C(s, t)) = V(G)$ . A  $k^*$ -laceable graph is a bipartite graph that there exists a  $k^*$ -container between every two vertices with different color. Thus, a graph is  $1^*$ -laceable and  $2^*$ -laceable if and only if it is Hamiltonian laceable. In [1], Chang et al. showed that  $Q_n$  is  $k^*$ -laceable for  $1 \leq k \leq n$ .

In [2], Chen et al. proposed some more general concepts. A  $k$ -fan  $A(s, T|G)$  of a bipartite graph  $G = (V_0 \cup V_1, E)$  is a set of  $k$  paths from  $s \in V_0$  and  $T = \{t_i\}$  for  $0 \leq i \leq k - 1$  and  $t_0 \in V_1$ , and  $t_j \in V_0$ , for  $1 \leq j \leq k - 1$ , such that every two paths of them share only the vertex  $s$ . Let  $V(A(s, T|G))$  be the set of vertices incident with some paths in  $A(s, T|G)$ . A  $k^*$ -fan  $A(s, T|G)$  is a  $k$ -fan with  $V(A(s, T|G)) = V(G)$ . A graph  $G$  is  $k^*$ -fanable, if  $\forall s \in V_0$  and  $\forall T = \{t_0, t_1, \dots, t_{k-1} | t_0 \in V_1 \text{ and } t_1, t_2, \dots, t_{k-1} \in V_0\}$ , there exists a  $k^*$ -fan. A graph  $G$  is  $f$  edges  $k^*$ -fanable if  $G - F$  is  $k^*$ -fanable  $\forall F \subset E(G)$  with  $|F| = f$ . Chen et al. proved that  $Q_n$  is  $f$  edges  $k^*$ -fanable for  $n \geq 3$ ,  $0 \leq f \leq n - 2$ ,  $1 \leq k + f \leq n$ .

In [3], Hung et al. studied the adjacent vertices fault tolerance Hamiltonian laceability for hypercube. A set of  $f$  pairs of adjacent vertex  $F$  is consisted of  $f$  pairs of adjacent vertices. We can denote  $F$  as  $\{a_1, b_1, a_2, b_2, \dots, a_f, b_f | a_i \in V_0, b_i \in V_1 \text{ and } (a_i, b_i) \in E \text{ for } 1 \leq i \leq f\}$ . A bipartite graph  $G = (V_0 \cup V_1, E)$  is  $f$ -adjacency  $k$  edges Hamil-

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tonian laceable if  $G - F$  is  $k$  edges Hamiltonian laceable  $\forall f$  pairs of adjacent vertex set  $F$  of  $G$ . In [3], the authors proved that  $Q_n$  is  $f$ -adjacency  $(n-2-f)$  edges Hamiltonian laceable for  $n \leq n-3$ .

In this paper, we will study adjacent vertex fault tolerance for  $k^*$ -fanability of hypercube. A bipartite graph  $G = (V_0 \cup V_1, E)$  is  $f$ -adjacency  $l$  edges  $k^*$ -fanable if  $G - F$  is  $l$  edges  $k^*$ -fanable for all  $f$  pairs of adjacent vertex set  $F$  of  $G$ . We will prove that  $Q_n$  is  $f$ -adjacency  $l$  edges  $(n-f-l)^*$ -fanable, for  $f+l \leq n-2$  and  $l \geq 1$ .

The rest of this paper is organized as follows. In Section 2, we present some previous results needed for our proof. Section 3 proves the adjacency vertex fault tolerance for  $k^*$ -fanability of hypercube. Finally, some conclusions are given in Section 4.

## 2 Preliminaries

In this section, we will present some results needed of our proof in Section 3. In [4], Park et al. proposed the following property, named as *property 2H*.

For  $Q_n = (V_0 \cup V_1, E)$ ,  $\forall s_1, s_2 \in V_0$  and  $t_1, t_2 \in V_1$ , there exist two disjoint paths  $P(s_1, t_1)$  and  $P(s_2, t_2)$  which cover all vertices of  $Q_n$ .

**Lemma 1** *The graph  $Q_n - F$  has property 2H,  $\forall F \subset E(Q_n)$ ,  $|F| = (n-3)$  for  $n \geq 3$ .*

Hung et al. showed the following theorem in [3].

**Theorem 1** *The hypercube  $Q_n$  is  $f$ -adjacency  $(n-2-f)$  edges Hamiltonian laceable for  $0 \leq f \leq (n-3)$ .*

The following theorem is proposed in [2].

**Theorem 2** *The  $Q_n$  is  $l$  edges  $k^*$ -fanable for  $n \geq 3$ ,  $0 \leq l \leq n-2$ , and  $1 \leq k \leq n-l$ .*

**Ohservaion 1** *IF a graph  $G$  is  $l$  edges Hamiltonian laceable then  $G$  is  $l'$  edges Hamiltonian laceable for  $0 \leq l' < l$ .*

**Ohservaion 2** *IF a graph  $G$  is  $f$ -adjacency  $l$  edges fanable then  $G$  is  $f$ -adjacency  $l'$  edges fanable for  $0 \leq l' < l$ .*

## 3 The $f$ -adjacency $l$ edges fanability of Hypercube

In this section, we will prove the adjacency vertex fault tolerance for fanability of hypercube.

**Theorem 3** *The graph  $Q_n$  is  $f$ -adjacency  $l$  edges  $(n-f-l)^*$ -fanable, for  $f+l \leq n-2$ ,  $l \geq 1$ .*

**Proof:** We will prove this theorem by induction on  $n$ . Since  $Q_n$  is  $l$  edges  $(n-l)^*$ -fanable, this theorem is true for  $f = 0$ . Applying **Theorem 1**, we can obtain that  $Q_4$  is 1-adjacency 1 edge Hamiltonian laceable. Thus,  $Q_4$  is 1-adjacency 1 edge  $2^*$ -fanable. Hence, theorem is true for  $n = 4$ . Since  $Q_n$  is  $f$ -adjacency  $(n-2-f)$  edges Hamiltonian laceable,  $Q_n$  is  $2^*$ -fanable for  $f+l = n-2$ . In the following, we will assume that  $f \geq 1$  and  $f+l \leq n-3$ .

Let  $k = n-f-l$ . By induction hypothesis, the two subgraphs  $Q_{n-1}^0$  and  $Q_{n-1}^1$  are  $f$ -adjacency  $l$  edges  $(k-1)^*$ -fanable for  $1 \leq f+l \leq n-3$ ,  $l \geq 1$ . Let  $s$  be a black node and  $T = \{t_i \mid t_0 \text{ is a white node and all } t_i \text{ are black nodes for } 1 \leq i \leq k-1\}$ . Let  $T^i = T \cap V(Q_{n-1}^i)$ . We denote the set of faulty nodes of  $Q_n$  by  $F$ . Let  $F^i = F \cap V(Q_{n-1}^i)$ ,  $f^i = |F^i|$  for  $i = 0, 1$ . We can denote the set of faulty edges of  $Q_n$  by  $L$ . Let  $L^i = L \cap E(Q_{n-1}^i)$ ,  $l^i = |L^i|$  for  $i = 0, 1$ . By symmetry of hypercube, we can assume every pair of adjacently faulty vertices and faulty edges are either in  $Q_{n-1}^0$  or  $Q_{n-1}^1$ .

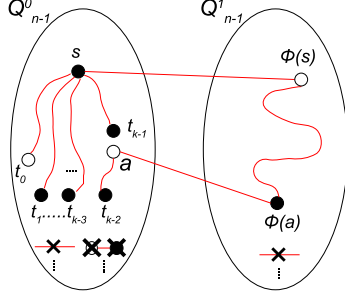
Without loss of generality, we can assume  $s \in V(Q_{n-1}^0)$ . We will construct  $k$  spanning disjoint paths between  $s$  and  $t_i$ , for  $0 \leq i \leq k-1$ ,  $f+l \leq n-3$ ,  $l \geq 1$  in following case.

**Case 1.**  $t_0 \in V(Q_{n-1}^0)$ .

**Case 1.1.**  $t_i \in V(Q_{n-1}^0)$ , for all  $1 \leq i \leq k-1$ , and  $f^1 = 0$ .

Since  $Q_{n-1}^0$  is  $f$ -adjacency  $l$  edges  $(k-1)^*$ -fanable, there exists a  $(k-1)^*$ -fan,  $A(s, \{t_0, \dots, t_{k-2}\} \mid Q_{n-1}^0 - F^0 - L^0)$ . We can write these  $k-1$   $(s, t_j)$ -paths as  $P_j^0$  for  $0 \leq j \leq k-2$ . Thus,  $t_{k-1}$  is on some path  $P_j^0$  for  $0 \leq j \leq k-2$ . Without loss of generality, we can assume  $t_{k-1}$  is on  $P_{k-2}^0$ . Thus, the path  $P_{k-2}^0$  can be written as  $\langle s \xrightarrow{P(s, t_{k-1})} t_{k-1}, a \xrightarrow{P(a, t_{k-2})} t_{k-2} \rangle$ . Applying **Theorem 1** we can construct a Hamiltonian path  $P(\phi(s), \phi(a))$  of  $Q_{n-1}^1$ . We can

construct the  $k^*$ -fan  $A(s, \{t_0, \dots, t_{k-1}\} \mid Q_n - F - L)$  as:  
 $P_j = P(s, t_j)$  for  $0 \leq j \leq k-3$ ,  
 $P_{k-2} = \langle s, \phi(s) \xrightarrow{P(\phi(s), \phi(a))} \phi(a),$   
 $a \xrightarrow{P(a, t_{k-2})} t_{k-2} \rangle$ ,  
 $P_{k-1} = P(s, t_{k-1})$ ,  
 as illustrated in Figure 1.


 Figure 1: Illustration of **Case 1.1**.

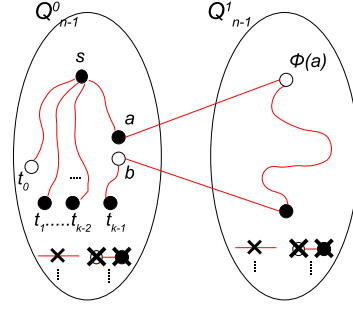
**Case 1.2.**  $t_i \in V(Q_{n-1}^0)$  for all  $1 \leq i \leq k-1$ , and  $f^1 > 0$ .

Since  $f^1 > 0$ ,  $Q_{n-1}^0 - F^0 - L^0$  is  $k^*$ -fanable. There exists a  $k^*$ -fan  $A(s, \{t_0, \dots, t_{k-1}\} \mid Q_{n-1}^0 - F^0 - L^0)$ . We can write these  $k$   $(s, t_j)$ -paths as  $P_j^0$  for  $0 \leq j \leq k-1$ . Let  $(a, b)$  be an edge on  $P_j^0$  such that  $\phi(a)$  and  $\phi(b)$  are fault-free for some  $0 \leq j \leq k-1$ . Without loss of generality, we can assume  $j = k-1$ . Thus, the path  $P_{k-1}^0$  can be written as  $\langle s \xrightarrow{P(s, a)} a, b \xrightarrow{P(b, t_{k-1})} t_{k-1} \rangle$ . Applying **Theorem 1** we can construct a Hamiltonian path  $P(\phi(a), \phi(b))$  in  $Q_{n-1}^1$ . We can construct the  $k^*$ -fan  $A(s, \{t_0, \dots, t_{k-1}\} \mid Q_n - F - L)$  as:

$P_j = P(s, t_j)$  for  $0 \leq j \leq k-2$ ,  
 $P_{k-1} = \langle s \xrightarrow{P(s, a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(b))} \phi(b), b \xrightarrow{P(b, t_{k-1})} t_{k-1} \rangle$ ,  
 as illustrated in Figure 2.

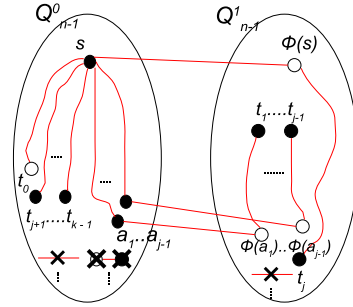
**Case 1.3.**  $t_i \in V(Q_{n-1}^1)$  for  $1 \leq i \leq j$ ,  $1 \leq j \leq k-1$ , and  $f^1 = 0$ .

Let  $E(t_j) = \{(t_j, \phi(v)) \mid \text{if } v \in F^0 \text{ or } v = t_r \text{ for } j+1 \leq r \leq k-1\}$ . Applying **Theorem 2**,  $Q_{n-1}^1$  is  $(k-1-j)$  edges  $j^*$ -fanable. Thus, there exists a  $j^*$ -fan  $A(t_j, \{\phi(s), t_1, \dots, t_{j-1}\} \mid Q_{n-1}^1 - E(t_j) - L^1)$ . These  $j$  paths can be written as  $P_0^1 = P(t_j, \phi(s))$ ,  $P_r^1 =$


 Figure 2: Illustration of **Case 1.2**.

$\langle t_j, \phi(a_r) \xrightarrow{P(\phi(a_r), t_r)} t_r \rangle$  for  $1 \leq r \leq j-1$ . Since  $Q_{n-1}^0$  is  $f$ -adjacency  $l$  edges  $(k-1)^*$ -fanable, there exists a  $(k-1)^*$ -fan  $A(s, \{t_0, a_1, \dots, a_{j-1}, t_{j+1}, \dots, t_{k-1}\} \mid Q_{n-1}^0 - F^0 - L^0)$ . We denote these  $k-1$  paths as:

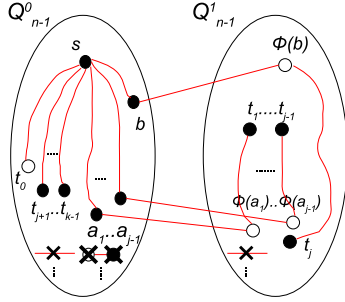
$P_0^0 = P(s, t_0)$ ,  
 $P_r^0 = P(s, a_r)$  for  $1 \leq r \leq j-1$ ,  
 $P_m^0 = P(s, t_m)$  for  $j+1 \leq m \leq k-1$ .  
 We can construct the  $(n-f-l)^*$ -fan  $A(s, \{t_0, \dots, t_{k-1}\} \mid Q_n - F - L)$  as:  
 $P_0 = P(s, t_0)$ ,  
 $P_r = \langle s \xrightarrow{P(s, a_r)} a_r, \phi(a_r) \xrightarrow{P(\phi(a_r), t_r)} t_r \rangle$  for  $1 \leq r \leq j-1$ ,  
 $P_j = \langle s, \phi(s) \xrightarrow{P(\phi(s), t_j)} t_j \rangle$ ,  
 $P_m = P(s, t_m)$  for  $j+1 \leq m \leq k-1$ ,  
 as illustrated in Figure 3.


 Figure 3: Illustration of **Case 1.3**.

**Case 1.4**  $t_i \in V(Q_{n-1}^1)$  for  $1 \leq i \leq j$ ,  $1 \leq j \leq k-1$ , and  $f^1 > 0$ .

Let  $b \in V(Q_{n-1}^0)$  be a black node such that  $b \notin T$  and  $\phi(b) \notin F$ . Let  $E(t_j) = \{(t_j, \phi(v)) \mid \text{if } v \in F \text{ or } v = t_m \text{ for } j+1 \leq m \leq k-1\}$ . Since  $Q_{n-1}^1 - F^1 - L^1$  is  $(k-1-j)$  edges  $j^*$ -fanable, there exists a

$j^*$ -fan  $A(t_j, \{\phi(b), t_1, \dots, t_{j-1}\} \mid Q_{n-1}^1 - F^1 - L^1 - E(t_j))$ . We can write these  $j$  paths as  $P_0^1 = P(t_j, \phi(b))$  and  $P_m^1 = \langle t_j, \phi(a_m) \xrightarrow{P(\phi(a_m), t_m)} t_m \rangle$  for  $1 \leq m \leq j-1$ . Since  $f^1 > 0$ ,  $Q_{n-1}^0$  is  $f^0$ -adjacency  $(l + f^1 - 1)$  edges  $k^*$ -fanable, there exists a  $k^*$ -fan  $A(s, \{t_0, a_1, \dots, a_{j-1}, t_{j+1}, \dots, t_{k-1}, b\} \mid Q_{n-1}^0 - F^0 - L^0)$ . These  $k$  paths can be written as:  
 $P_0^0 = P(s, t_0)$ ,  
 $P_m^0 = P(s, a_m)$  for  $1 \leq m \leq j-1$ ,  
 $P_j^0 = P(s, b)$ ,  
 $P_w^0 = P(s, t_w)$  for  $j+1 \leq w \leq k-1$ .  
 We can construct the  $k^*$ -fan  $A(s, \{t_0, \dots, t_{k-1}\} \mid Q_n - F - L)$  as:  
 $P_0 = P(s, t_0)$ ,  
 $P_m = \langle s \xrightarrow{P(s, a_m)} a_m, \phi(a_m) \xrightarrow{P(\phi(a_m), t_m)} t_m \rangle$  for  $1 \leq m \leq j-1$ ,  
 $P_j = \langle s \xrightarrow{P(s, b)} b, \phi(b) \xrightarrow{P(\phi(b), t_j)} t_j \rangle$ ,  
 $P_w = P(s, t_w)$  for  $j+1 \leq w \leq k-1$ ,  
 as illustrated in Figure 4.


 Figure 4: Illustration of **Case 1.4**.

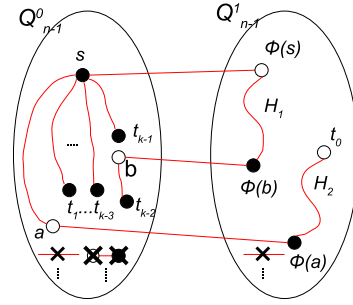
**Case 2.**  $t_0 \in V(Q_{n-1}^1)$ .

**Case 2.1.**  $t_i \in V(Q_{n-1}^0)$ , for all  $1 \leq i \leq k-1$ , and  $f^1 = 0$ .

Let  $a$  be a white node in  $Q_{n-1}^0$ . Since  $Q_{n-1}^0$  is  $f$ -adjacency  $l$  edges  $(k-1)^*$ -fanable, there exists a  $(k-1)^*$ -fan  $A(s, \{a, t_1, \dots, t_{k-2}\} \mid Q_{n-1}^0 - F^0 - L^0)$ . We can write these  $k-1$  paths as  $P_0^0 = P(s, a)$  and  $P_m^0 = P(s, t_m)$  for  $1 \leq m \leq k-2$ . Thus,  $t_{k-1}$  is on some paths  $P_m^0$  for  $0 \leq m \leq k-2$ . Without loss of generality, we can assume  $t_{k-1}$  is on  $P_{k-2}^0$ . Thus,  $P_{k-2}^0$  can be written as  $\langle s \xrightarrow{P(s, t_{k-1})} t_{k-1}, b \xrightarrow{P(b, t_{k-2})} t_{k-2} \rangle$ . Suppose that

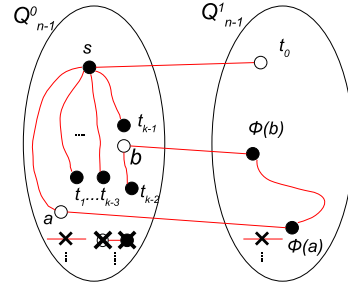
$\phi(s) \neq t_0$ . Applying **Lemma 1**, we can obtain that there exist two spanning disjoint paths  $H_1$  (joining  $\phi(s)$  and  $\phi(b)$ ), and  $H_2$  (joining  $\phi(a)$  and  $t_0$ ). We can construct the  $k^*$ -fan  $A(s, \{t_0, \dots, t_{k-1}\} \mid Q_n - F - L)$  as:

$P_0 = \langle s \xrightarrow{P(s, a)} a, \phi(a) \xrightarrow{H_2} t_0 \rangle$ ,  
 $P_m = P(s, t_m)$  for  $1 \leq m \leq k-3$ ,  
 $P_{k-2} = \langle s, \phi(s) \xrightarrow{H_1} \phi(b), b \xrightarrow{P(b, t_{k-2})} t_{k-2} \rangle$ ,  
 $P_{k-1} = P(s, t_{k-1})$ ,  
 as illustrated in Figure 5.


 Figure 5: Illustration of **Case 2.1.1**.

Suppose that  $\phi(s) = t_0$ . There exists a Hamiltonian path  $P(\phi(a), \phi(b))$  in  $Q_{n-1}^1 - \{t_0\} - L^1$  since  $Q_n$  is  $(n-3)$  edge hyper-Hamiltonian laceable. We can construct the  $k^*$ -fan  $A(s, \{t_0, \dots, t_{k-1}\} \mid Q_n - F - L)$  as:

$P_0 = \langle s, t_0 \rangle$ ,  
 $P_m = P(s, t_m)$  for  $1 \leq m \leq k-3$ ,  
 $P_{k-2} = \langle s \xrightarrow{P(s, a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(b))} \phi(b), b \xrightarrow{P(b, t_{k-2})} t_{k-2} \rangle$ ,  
 $P_{k-1} = P(s, t_{k-1})$ ,  
 as illustrated in Figure 6.


 Figure 6: Illustration of **Case 2.1.2**.

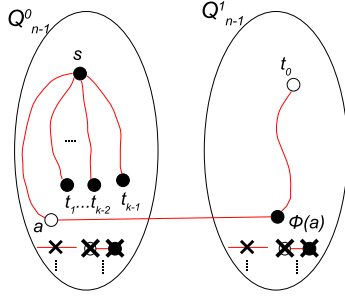
**Case 2.2.**  $t_i \in V(Q_{n-1}^0)$  for all  $1 \leq i \leq k-1$ , and  $f^1 > 0$ .

Let  $a$  be a white node in  $Q_{n-1}^0$  such that  $a, \phi(a) \notin F$ . Since  $f^1 > 0$ ,  $Q_{n-1}^0$  is  $f^0$ -adjacency  $(l + f^1 - 1)$  edges  $k^*$ -fanable, there exists a  $k^*$ -fan  $A(s, \{a, t_1, \dots, t_{k-1}\} \mid Q_{n-1}^0 - F^0 - L^0)$ . We can write these  $k$  paths as  $P_0^0 = P(s, a)$  and  $P_m^0 = P(s, t_m)$  for  $1 \leq m \leq k - 1$ . Applying **Theorem 1**, there exists a Hamiltonian path  $P(\phi(a), t_0)$  in  $Q_{n-1}^1 - F^1 - L^1$ . We can construct the  $k^*$ -fan  $A(s, \{t_0, t_1, \dots, t_{k-1}\} \mid Q_n - F - L)$  as:

$$P_0 = \langle s \xrightarrow{P(s,a)} a, \phi(a) \xrightarrow{P(\phi(a),t_0)} t_0 \rangle,$$

$$P_m = P(s, t_m), \text{ for } 1 \leq m \leq k - 1,$$

as illustrated in Figure 7.


 Figure 7: Illustration of **Case 2.2**.

**Case 2.3.**  $t_i \in V(Q_{n-1}^1)$  for  $1 \leq i \leq j$ ,  $1 \leq j \leq k - 1$ , and  $f^1 = 0$ .

Since  $Q_{n-1}^1$  is  $f$ -adjacency  $l$  edges  $(k - 1)^*$ -fanable. Let  $\phi(a_0)$  is a neighbor of  $t_0$  in  $Q_{n-1}^1$  and  $a_0 \notin F$ .  $Q_{n-1}^1 - L^1 - \{t_0, \phi(a)\}$  is  $(k - 2 - j)$  edges  $j^*$ -fanable. Let  $E(t_j) = \{(t_j, \phi(v)) \mid \text{if } v \in F \text{ or } v = t_m \text{ for } j + 1 \leq m \leq k - 1\}$ . Thus, there exists a  $j^*$ -fan  $A(t_j, \{\phi(s), t_1, \dots, t_{j-1}\} \mid Q_{n-1}^1 - L^1 - \{t_0, \phi(a)\} - E(t_j))$ . We can write these  $j$  paths as  $P_0^1 = P(t_j, \phi(s))$  and  $P_m^1 = \langle t_j, \phi(a_m) \xrightarrow{P(\phi(a_m), t_m)} t_m \rangle$  for  $1 \leq m \leq j - 1$ . Since  $Q_{n-1}^0$  is  $f$ -adjacency  $l$  edges  $(k - 1)^*$ -fanable, there exists a  $(k - 1)^*$ -fan  $A(s, \{a_0, a_1, \dots, a_{j-1}, t_{j+1}, \dots, t_{k-1}\} \mid Q_{n-1}^0 - F^0 - L^0)$ .

$$P_m^0 = P(s, a_m) \text{ for } 0 \leq m \leq j - 1,$$

$$P_r^0 = P(s, t_r) \text{ for } j + 1 \leq r \leq k - 1.$$

We can construct the  $k^*$ -fan  $F(s, t_0, \dots, t_{k-1} \mid Q_n - F - L)$  as:

$$P_0 = \langle s \xrightarrow{P(s,a_0)} a_0, \phi(a_0), t_0 \rangle,$$

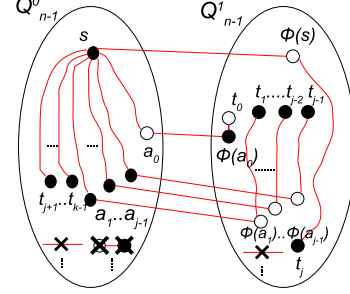
$$P_m = \langle s \xrightarrow{P(s,a_m)} a_m, \phi(a_m) \xrightarrow{P(\phi(a_m), t_m)} t_m \rangle$$

$t_m \rangle$  for  $1 \leq m \leq j - 1$ ,

$$P_j = \langle s, \phi(s) \xrightarrow{P(\phi(s), t_j)} t_j \rangle,$$

$$P_r = P(s, t_r) \text{ for } j + 1 \leq r \leq k - 1,$$

as illustrated in Figure 8.


 Figure 8: Illustration of **Case 2.3**.

**Case 2.4.**  $t_i \in V(Q_{n-1}^1)$  for  $1 \leq i \leq j$ ,  $1 \leq j \leq k - 1$ , and  $f^1 > 0$ .

Suppose that there exists a vertex  $t_i = \phi(b)$  for some  $1 \leq i \leq j$  such that  $b$  is fault-free. Without loss of generality, we can assume  $i = j$ . Let  $E(t_j) = \{(t_j, \phi(v)) \mid \text{if } v \in F \text{ or } v = t_m \text{ for } j + 1 \leq m \leq k - 1\}$ . Since  $Q_{n-1}^1 - F^1 - L^1$  is  $(k - 1 - j)$  edge  $j^*$ -fanable, there exists a  $j^*$ -fan  $A(t_j, \{t_0, t_1, \dots, t_{j-1}\} \mid Q_{n-1}^1 - F^1 - L^1 - E(t_j))$ . We can write these  $j$  paths

$$\text{as } P_m^1 = \langle t_j, \phi(a_m) \xrightarrow{P(\phi(a_m), t_m)} t_m \rangle$$

for  $0 \leq m \leq j - 1$ . Since  $f^1 > 0$ ,  $Q_{n-1}^0$  is  $f^0$ -adjacency  $(l + f^1 - 1)$  edges  $k^*$ -fanable, there exists a  $k^*$ -fan  $A(s, \{b, a_0, \dots, a_{j-1}, t_{j+1}, \dots, t_{k-1}\} \mid Q_{n-1}^0 - F^0 - L^0)$ . We can write these  $k$  paths  $P_0^0 = P(s, b)$ ,  $P_m^0 = P(s, a_m)$ , for  $0 \leq m \leq j - 1$ ,  $P_r^0 = P(s, t_r)$ , for  $j + 1 \leq r \leq k - 1$ . We can construct the  $k^*$ -fan  $A(s, \{t_0, \dots, t_{k-1}\} \mid Q_n - F - L)$  as:

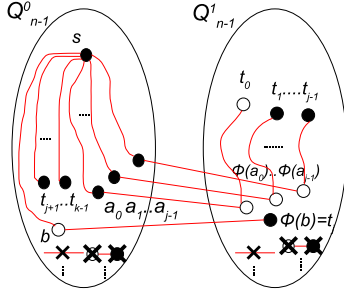
$$P_m = \langle s \xrightarrow{P(s,a_m)} a_m, \phi(a_m) \xrightarrow{P(\phi(a_m), t_m)} t_m \rangle \text{ for } 0 \leq m \leq j - 1,$$

$$P_j = \langle s \xrightarrow{P(s,b)} b, t_j \rangle,$$

$$P_r = P(s, t_r) \text{ for } j + 1 \leq r \leq k - 1,$$

as illustrated in Figure 9.

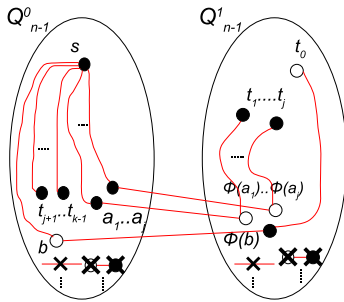
Otherwise, every neighbor of  $t_i$  for  $1 \leq i \leq j$  in  $Q_{n-1}^0$  are faulty. Since  $2^{n-2} > n - 3$  for  $n \geq 5$ , there exists a white vertex  $b \in V(Q_{n-1}^0) - F^0$  such that  $\phi(b)$  is a black non-faulty vertex. Let  $E(\phi(b)) = \{(\phi(b), \phi(v)) \mid \text{if } v \in F \text{ or } v = t_m \text{ for } j + 1 \leq m \leq k - 1\}$ .


 Figure 9: Illustration of **Case 2.4.1.**

Since  $Q_{n-1}^1 - F^1 - L^1$  is  $(k - 2 - j)$  edges  $(j + 1)^*$ -fanable. There exists a  $(j + 1)^*$ -fan  $A(\phi(b), \{t_0, \dots, t_j\} \mid Q_{n-1}^1 - F^1 - L^1 - E(\phi(b)))$ . We can write these  $j$  paths as  $P_0^1 = P(\phi(b), t_0)$  and  $P_m^1 = \langle \phi(b), \phi(a_m) \xrightarrow{P(\phi(a_m), t_m)} t_m \rangle$  for  $1 \leq m \leq j$ . Since  $f^1 > 0$ ,  $Q_{n-1}^0$  is  $f$ -adjacency  $l$  edges  $k^*$ -fanable, there exists a  $k^*$ -fan  $A(s, \{b, a_1, \dots, a_j, t_{j+1}, \dots, t_{k-1}\} \mid Q_{n-1}^0 - F^0 - L^0)$ . We can write these  $k$  paths  $P_0^0 = P(s, b)$ ,  $P_m^0 = P(s, a_m)$  for  $1 \leq m \leq j$ ,  $P_r^0 = P(s, t_r)$  for  $j + 1 \leq r \leq k - 1$ .

We can construct the  $k^*$ -fan  $A(s, \{t_0, \dots, t_{k-1}\} \mid Q_n - F - L)$  as:

$P_0 = \langle s \xrightarrow{P(s,b)} b, \phi(b) \xrightarrow{P(\phi(b), t_0)} t_0 \rangle$ ,  
 $P_m = \langle s \xrightarrow{P(s, a_m)} a_m, \phi(a_m) \xrightarrow{P(\phi(a_m), t_m)} t_m \rangle$  for  $1 \leq m \leq j$ ,  
 $P_r = P(s, t_r)$  for  $j + 1 \leq r \leq k - 1$ ,  
 as illustrated in Figure 10.


 Figure 10: Illustration of **Case 2.4.2.**

This theorem is proved.

If  $l = 0$ ,  $Q_n$  is  $f$ -adjacency  $(n - 1 - f)^*$ -fanable for  $0 \leq f \leq n - 3$ ,  $n \geq 3$ .

**Corollary 1** The graph  $Q_n$  is  $f$ -adjacency  $l$  edges  $k^*$ -fanable for  $1 \leq k \leq n - f - l$ ,  $f + l \leq n - 2$ ,  $l \geq 1$ .

## 4 Conclusions

In this paper, we show that the  $n$ -dimensional hypercube  $Q_n$  is  $f$ -adjacency  $l$  edges  $k^*$ -fanable for  $1 \leq k \leq n - f - l$ ,  $f + l \leq n - 2$ ,  $l \geq 1$ . It is worth while to investigate more vertex fault tolerance for fanability of hypercube networks.

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